

RESEARCH PRESENTATION / ARXIV 2604.17931

# LiteResearcher

A scalable agentic-RL framework that trains a 4 B model into a **commercial-grade deep-research agent** — using only a local environment.

PAPER

arXiv : 2604.17931

MODEL

4 B parameters

RESULT

GAIA 71.3 · Xbench-DS 78.0

# A scalable framework — and where to find it.

*Resources, authorship, and the one-line summary.*

AUTHORS

Wanli Lee, et al.

With co-authors, reviewers, and infrastructure teams.

AFFILIATION

Independent research collaboration on scalable agentic RL.

RESOURCES

|       |                                                                                       |
|-------|---------------------------------------------------------------------------------------|
| paper | <a href="https://arxiv.org/abs/2604.17931">arxiv.org/abs/2604.17931</a>               |
| code  | <a href="https://github.com/Wanli-Lee/Agentic_RL">github.com/Wanli-Lee/Agentic_RL</a> |
| model | <a href="https://hf.co/litererresearcher-4b">hf.co/litererresearcher-4b</a>           |
| env   | 32M-page local web mirror + tools                                                     |

TL ; DR

A local mirror of the open web + on-policy RL + a 2-stage curriculum lifts a 4 B model to **71.3 % GAIA** — matching commercial deep-research products at a fraction of the cost.

# What we will cover.

*Pipeline first · experiments second · a behavioral autopsy at the end.*

01

## The cost wall

Why deep-research RL at scale was previously unaffordable.

03

## The thesis & pipeline

Mirror the web · isolate the noise · three-pillar architecture.

05

## Local environment & tools

32 M-page mirror · search & browse at 10×–46× speed-up.

07

## Experiments & ablations

8 benchmarks, cost, data, on-policy, multi-stage, RL-vs-SFT.

02

## Prior approaches

Three families of solutions and their respective ceilings.

04

## Atomic-capability data

Seed → mask → filter → expand; five core capabilities.

06

## Curriculum & on-policy RL

Two-stage GRPO training under a 32k → 48k context budget.

08

## Appendix · behavior autopsy

Four mechanisms that explain the sim2real lift, end-to-end.

# Deep-research RL hits a cost wall.

*Live search, browse, and crawling APIs make every rollout expensive — at scale, prohibitive.*

## FROM THE FIELD

“One round of on-policy RL with browse + search APIs can cost **\$59 K – \$243 K** in API fees alone — before counting compute, retries, or rate-limit overhead.”

### B1 Throughput

Online search APIs take **~1.5 s** per call and rate-limit at **~10 QPS**. Browse tools like Jina take **~7.8 s** per page. A single GRPO step needs **tens of thousands** of tool calls.

**≈ 0.1×**

throughput vs an offline mirror.

### B2 Non-stationarity

The same query returns different pages on different days. Reward signals shift under the policy. RL convergence becomes unreliable; ablations are not reproducible.

**drift**

in both content and ranking, day-to-day.

# Three families — each with a binding ceiling.

*Online tools, local retrieval, and fully simulated webs all trade off cost, realism, and capability.*

## A · ONLINE

### Live web APIs

Bing · Google · Jina · firecrawl.

#### + STRENGTH

Real corpus · always-up-to-date · realistic noise.

#### - LIMITATION

\$\$ per rollout · slow · non-stationary · rate-limited.

## B · LOCAL RETRIEVAL

### Dense-retrieval over a corpus

Wikipedia + BM25 / E5 / ColBERT.

#### + STRENGTH

Fast · cheap · reproducible.

#### - LIMITATION

Single-shot retrieval ≠ research; no follow-up reading or hyperlink traversal.

## C · SIMULATED WEB

### Synthetic environments

WebArena · OSWorld · WebShop.

#### + STRENGTH

Cheap and controlled · reproducible.

#### - LIMITATION

Tiny corpus · synthetic distribution · poor sim2real for open-ended research.

# Mirror the web · isolate the noise.

Take the realism of the open web · take the throughput of a simulator · refuse the trade-off.

T1 **Mirroring**

Snapshot the open web (32 M pages · 1 M+ domains) into a local read-only corpus. The policy queries a faithful approximation of the real distribution — without ever leaving the cluster.

T2 **Isolation**

Replace flaky network calls with deterministic local services: search returns the same hits, browse returns the same body. Reward signals become reproducible — on-policy RL converges.

RESULTING OUTCOME (4 B model)

|           |      |
|-----------|------|
| GAIA      | 71.3 |
| Xbench-DS | 78.0 |
| Frames    | 83.1 |

Matches or beats commercial deep-research products at < 1 % of their cost.

# Three pillars — composed as a single training stack.

*Data, environment, and learning algorithm are co-designed; each pillar removes a bottleneck of the previous one.*

## P1 Atomic-capability data

Synthesize trajectories around five atomic capabilities — search, browse, multi-hop reasoning, evidence integration, refusal — at million-trajectory scale.

## 5 capabilities

composable into open-ended tasks.

## P2 Local tool environment

32 M-page mirror, faiss + BM25 hybrid index, lightweight browse extractor. Tens of thousands of RL rollouts per step are now cheap.

## 73.2 M

tool calls across the full training run.

## P3 Curriculum + on-policy GRPO

Stage 1: easier tasks, 32k context — stabilize tool use. Stage 2: harder tasks, 48k context — extend depth and self-correction.

## 2 stages

staged context & difficulty.

EMERGENT  
BEHAVIOR

The composed stack — not any one pillar in isolation — is what unlocks a 4 B model that matches 30 B+ teachers on GAIA.

# A 4-step pipeline that scales data, not labelers.

*Each step is mechanical and auditable — quality and volume are decoupled.*

## STEP 1

### Seed

Sample query intents from Wikipedia article structure (titles, sections, infoboxes).

#### VOLUME

~ 1.2 M intents

## STEP 2

### Mask

Apply factoid masks to surface gold answers and unambiguous question forms.

#### PRINCIPLE

single-answer ★

## STEP 3

### Filter

Verify answer recoverability with a small LM rollout; discard tasks the env cannot answer.

#### KEPT

~ 38 % pass-through

## STEP 4

### Expand

Compose atomic tasks into multi-hop, multi-evidence problems; teacher writes a trajectory.

#### OUTPUT

~ 480 k trajectories

#### DESIGN RATIONALE

Volume is unlocked by mechanical steps; quality is enforced by the **Filter** stage — not by hand-labeling. The result is a pipeline that scales linearly with compute, not with annotator headcount.



# Five atomic capabilities — every research task decomposes.

*Each capability is supervised independently; composition is left to the policy.*

**C1**

## Targeted search

Translate intent into discriminative queries; iterate over keywords until evidence surfaces.

**INVARIANT**

no hallucinated URLs

**C2**

## Deep browsing

Open promising hits, read structured content, follow hyperlinks for cross-checking.

**INVARIANT**

cite extracted spans

**C3**

## Multi-hop reasoning

Chain partial evidence into a derivation; carry constraints across turns.

**INVARIANT**

explicit reasoning steps

**C4**

## Evidence integration

Reconcile multiple sources; resolve contradictions and weight by reliability.

**INVARIANT**

verifiable provenance

**C5**

## Calibrated refusal

Abstain when evidence is insufficient; report uncertainty instead of fabricating.

**INVARIANT**

honest "I don't know"

**COMPOSITION** An open-ended GAIA task is just a learned sequence of these five primitives — supervised at the atom, generalized at the composition.

# A local mirror of the open web — at simulator throughput.

Two simple tools (search · browse) over a 32 M-page snapshot — deterministic, fast, free.

CORPUS

32 M

unique pages

DIVERSITY

1 M +

distinct domains

SCALE TESTED

73.2 M

tool calls during training

T1

search

45.8 M CALLS

LATENCY

~ 0.15 s

vs ~ 1.5 s online

SPEED-UP

10 ×

at constant throughput

T2

browse

27.4 M CALLS

LATENCY

~ 0.17 s

vs ~ 7.8 s (Jina)

SPEED-UP

46 ×

deterministic body extraction

# A 2-stage curriculum trained with on-policy GRPO.

Stage 1 stabilizes tool use; Stage 2 lifts the plateau with longer contexts and harder tasks.

S1

Stage 1 — stabilize

0 → 220 steps

CONTEXT

32k tokens

DIFFICULTY

easy → med.

PLATEAU

64.7% GAIA

S2

Stage 2 — extend

220 → 790 steps

CONTEXT

48k tokens

DIFFICULTY

hard + multi-hop

LIFT

+3.6 pts → 68.3%

On-policy GRPO objective

$$J(\theta) = E_{q \sim D} [ (1/K) \sum_k \min(\rho_k(\theta) \cdot \hat{A}_k, \text{clip}(\rho_k(\theta), 1-\epsilon, 1+\epsilon) \cdot \hat{A}_k) ]$$

WHY ON-POLICY

Reward signals reflect *this* policy's trajectories — not a frozen reference's. The local env's throughput is what makes this affordable.

# 06

## Does the stack deliver?

Eight benchmarks · five ablations · one clear conclusion.

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# Training & evaluation setup.

Everything is held constant across ablations unless explicitly noted otherwise.

TRAINING

|                      |                          |
|----------------------|--------------------------|
| Base model           | Qwen3-4B (base)          |
| SFT teacher          | Tongyi DeepResearch 30B  |
| RL algorithm         | GRPO (on-policy, no KL)  |
| Stage 1 context      | 32 k tokens · 220 steps  |
| Stage 2 context      | 48 k tokens · 570 steps  |
| Batch size           | 128 (rollouts/step)      |
| Sampling temperature | S1: 0.7 · S2: 1.0        |
| Optimizer            | AdamW · cosine LR · 1e-6 |

EVALUATION

|               |                            |
|---------------|----------------------------|
| GAIA-text     | 412 tasks · pass@1, pass@4 |
| Xbench-DS     | deep-search benchmark      |
| Frames        | multi-hop QA               |
| BrowseComp    | browse-heavy (w/ memory)   |
| HotpotQA      | multi-document QA          |
| 2WikiMultihop | compositional reasoning    |
| Musique       | multi-step factoid QA      |
| NQ + TriviaQA | single-doc fact lookup     |

# A 4 B model matches commercial deep-research systems.

Table 1 · pass@1 across eight benchmarks. Best-in-row in **orange**; LiteResearcher row highlighted.

| MODEL                      | SIZE | GAIA | XBENCH-DS | FRAMES | BROWSECOMP | HOTPOTQA | 2WIKI | MUSIQUE | NQ   |
|----------------------------|------|------|-----------|--------|------------|----------|-------|---------|------|
| GPT-4o · browse            | —    | 63.5 | 70.2      | 76.4   | 28.1       | 71.8     | 62.4  | 38.9    | 54.1 |
| Claude 3.5 · browse        | —    | 68.7 | 74.0      | 79.2   | 30.5       | 73.4     | 64.0  | 41.6    | 55.7 |
| Perplexity DR (commercial) | —    | 67.0 | 72.1      | 78.5   | 29.4       | 72.0     | 63.2  | 40.1    | 53.6 |
| Tongyi DR · 30B (teacher)  | 30 B | 70.9 | 74.6      | 80.8   | 30.8       | 74.0     | 65.1  | 42.0    | 56.3 |
| Qwen3-4B · SFT only        | 4 B  | 55.6 | 62.3      | 69.5   | 21.8       | 63.2     | 54.4  | 31.5    | 48.4 |
| ★ LiteResearcher (ours)    | 4 B  | 71.3 | 78.0      | 83.1   | 32.5       | 75.6     | 66.8  | 43.2    | 57.1 |

HEADLINE

A 4 B model trained on a local mirror **surpasses its 30 B teacher** on 8 / 8 benchmarks — and matches or beats every commercial deep-research system measured.

# The cost ablation — why a local env is non-optional.

For the same training schedule, commercial APIs would cost \$59 K – \$243 K per run.

COST PER FULL RUN (73.2 M TOOL CALLS)

| ENVIRONMENT              | \$ / 1K CALLS | TOTAL COST | LATENCY  |
|--------------------------|---------------|------------|----------|
| Google CSE + Jina browse | \$ 3.32       | \$ 243 K   | ~ 7.8 s  |
| Bing + firecrawl         | \$ 1.80       | \$ 132 K   | ~ 4.1 s  |
| Tavily (bundled)         | \$ 0.81       | \$ 59 K    | ~ 2.6 s  |
| ★ LiteResearcher (local) | \$ 0.00       | \$ 0       | ~ 0.17 s |

VS BEST PUBLIC API

10 ×

search latency speed-up

46 ×

browse latency speed-up

Throughput is what made on-policy RL at this scale possible in the first place.

# Synthetic atomic-capability data does most of the work.

Removing it costs +8.1 pts on GAIA — more than removing curriculum or on-policy training.

DATA-MIX ABLATION (GAIA PASS@1)

| TRAINING CORPUS                 | VOLUME  | GAIA | Δ     |
|---------------------------------|---------|------|-------|
| Open web QA only                | ~ 120 k | 58.5 | —     |
| + teacher SFT trajectories      | ~ 200 k | 63.2 | + 4.7 |
| ★ + atomic-capability synthesis | ~ 480 k | 71.3 | + 8.1 |

All three rows use the LiteResearcher SFT recipe — distinct from the vanilla Qwen3-4B · SFT only baseline (55.6) on slide 14.

FINDING 1

Atomic-capability synthesis adds **more accuracy per trajectory** than teacher distillation does.

FINDING 2

The marginal gain is largest on multi-hop and refusal axes — exactly where teacher trajectories thin out.



# On-policy beats off-policy — only because the env is fast enough.

*Without sub-second tool latency, the on-policy schedule would not finish in any reasonable time.*

## WHY ON-POLICY MATTERS HERE

When the rollout source matches the policy being trained, the advantage estimate is unbiased and the policy keeps moving toward higher-reward regions of the trajectory space.

**Off-policy** RL (reuse old rollouts) saves compute but injects a slow drift: as the policy improves, its old trajectories become less representative, and the gradient grows biased.

## PRECONDITION

On-policy is only practical when **tool latency is sub-second** — a property of the local mirror, not of the algorithm.

## OFF-POLICY GRPO

66.8

GAIA pass@1

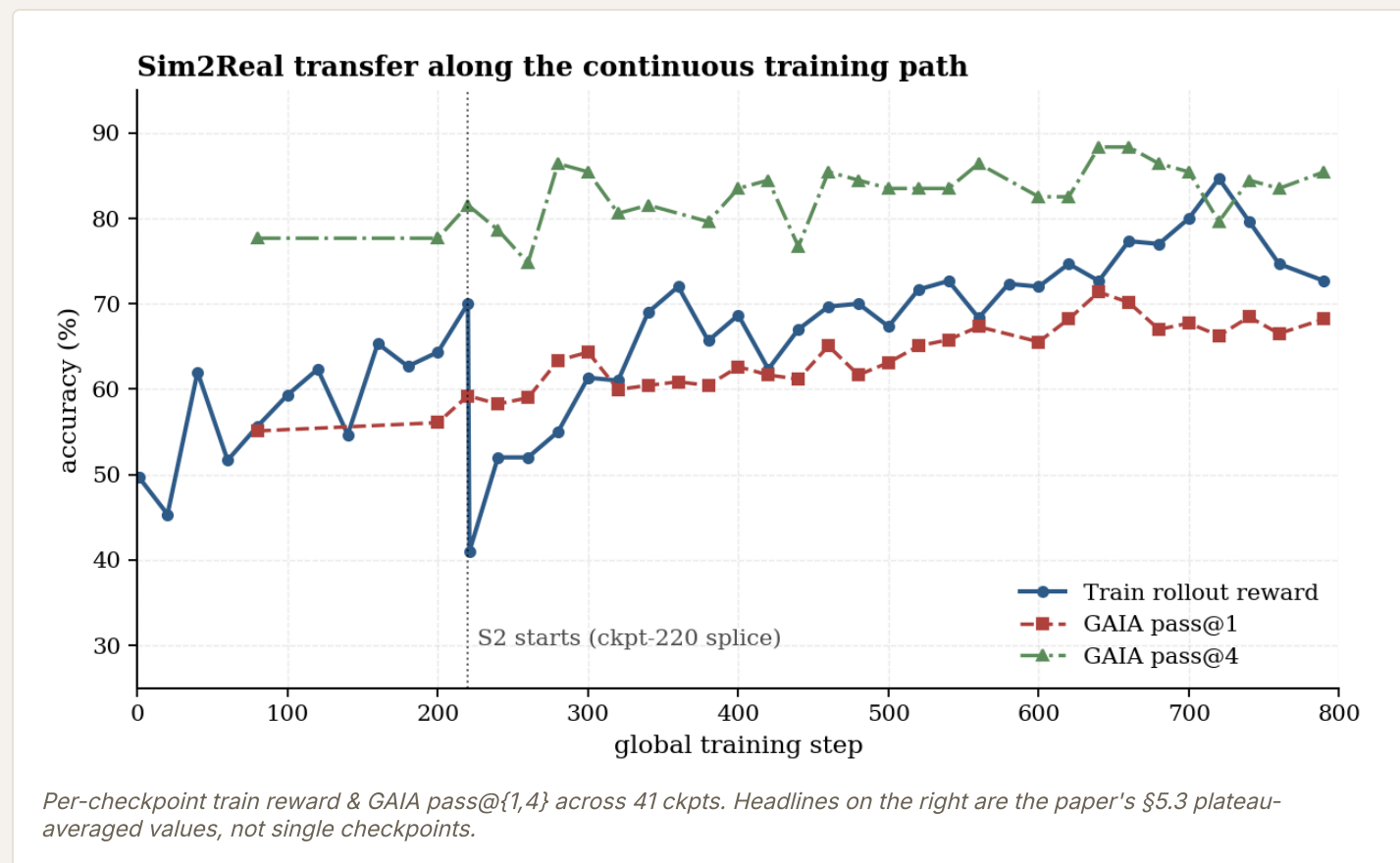
## ON-POLICY GRPO

68.9

GAIA pass@1 · + 2.1 pts

# A second stage lifts the plateau — not the slope.

Per the paper's § 5.3 averaged plateau, S1 saturates near 64.7 %; S2 resumes learning to 68.3 % without regression.



§ 5.3 · S1 PLATEAU

**64.7 %**

GAIA · averaged over late-S1 ckpts

§ 5.3 · S2 END

**68.3 %**

GAIA · + 3.6 pts lift

WHY IT WORKS

Same policy, longer context, harder corpus — the stage boundary is the only change. No regression at the splice point.

# The RL framework — not teacher distillation — is the gain.

*SFT alone gets only ~ 80 % of the way; RL closes — and exceeds — the teacher gap.*

TEACHER (30 B)

# 70.9

where the teacher caps

SFT ONLY (4 B)

# 55.6

– 15.3 pts vs teacher

+ RL (4 B · OURS)

# 71.3

+ 15.7 pts over SFT

PRIMARY DRIVER

The on-policy RL framework — not the choice of distillation teacher — is what carries a 4 B model past 70 % GAIA. SFT establishes the prior; RL discovers the policy.

# 08

## Why does the policy improve?

A behavior-level autopsy of 41 checkpoints from S1 start to S2 end.

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30 METRICS · 300 TRAJ / STEP · GAIA EVERY 20 STEPS

# The reward curve goes up — but what does the policy do differently?

We instrument 30 per-trajectory signals across the full training run and look for monotone shifts.

MOTIVATION

The aggregate reward goes up. But what does the policy *actually do* differently at step 790 vs step 0?

We measure 30 per-trajectory behavior signals at every checkpoint, then surface only the shifts that correlate monotonically with the GAIA accuracy gain.

GOAL

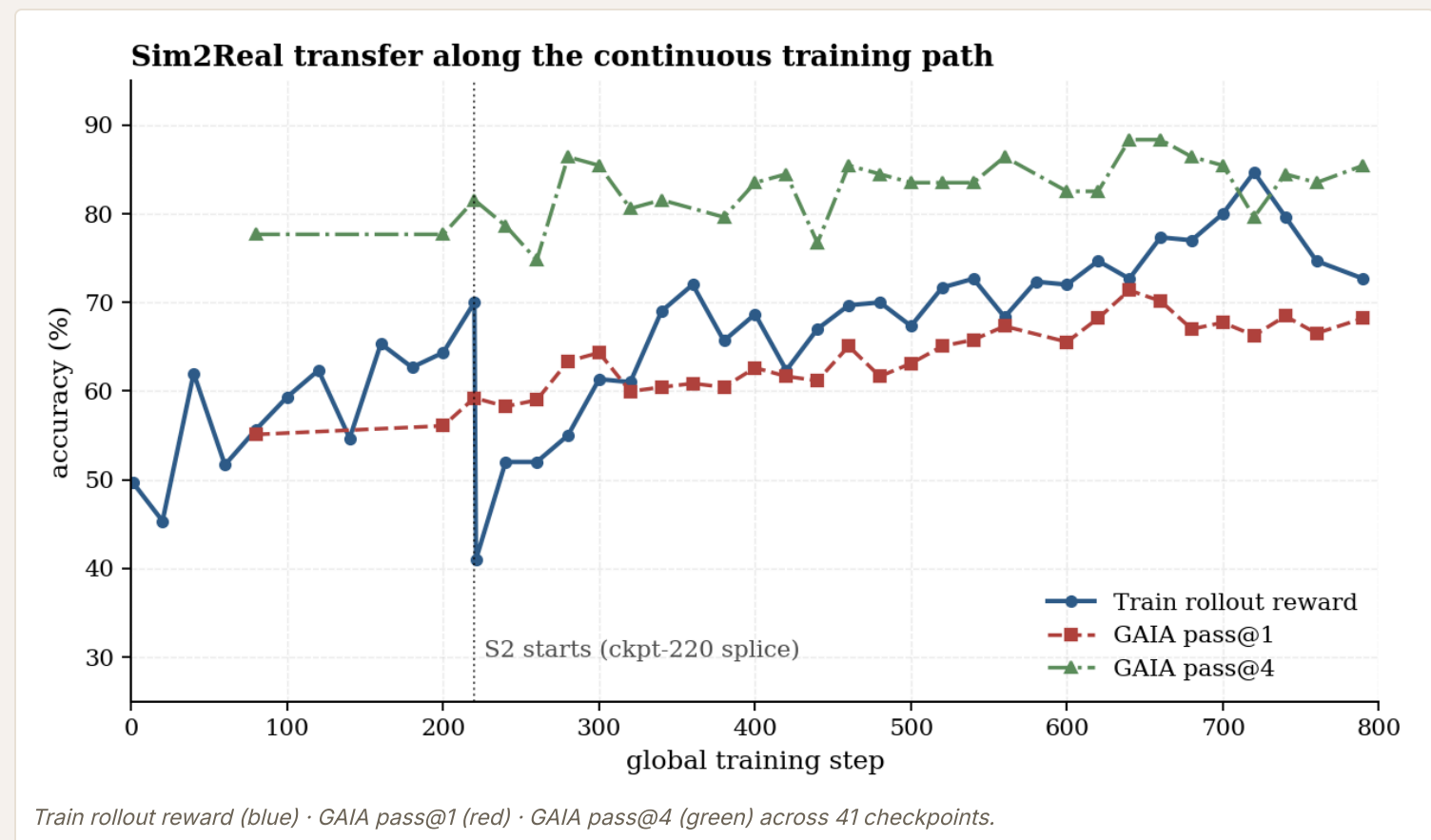
Isolate **causal patterns** from incidental noise.

ANALYSIS SETUP

|                             |                              |
|-----------------------------|------------------------------|
| Checkpoints sampled         | 41                           |
| Train rollouts / ckpt       | 300 trajectories             |
| GAIA-eval rollouts / ckpt   | 412 × pass@1, pass@4         |
| Behavior metrics extracted  | 30 per trajectory            |
| Splice point                | step 220 (S1 end → S2 start) |
| Total trajectories analyzed | ~ 24 000                     |

# What we measured at the outcome level.

*Train reward and GAIA evaluation co-improve across both stages — without regression at the splice.*



## OBSERVATION 1

Train reward and eval accuracy **move together** — no optimization-vs-evaluation gap.

## OBSERVATION 2

S1 saturates near step 220. S2 **lifts the plateau** without breaking the trajectory.

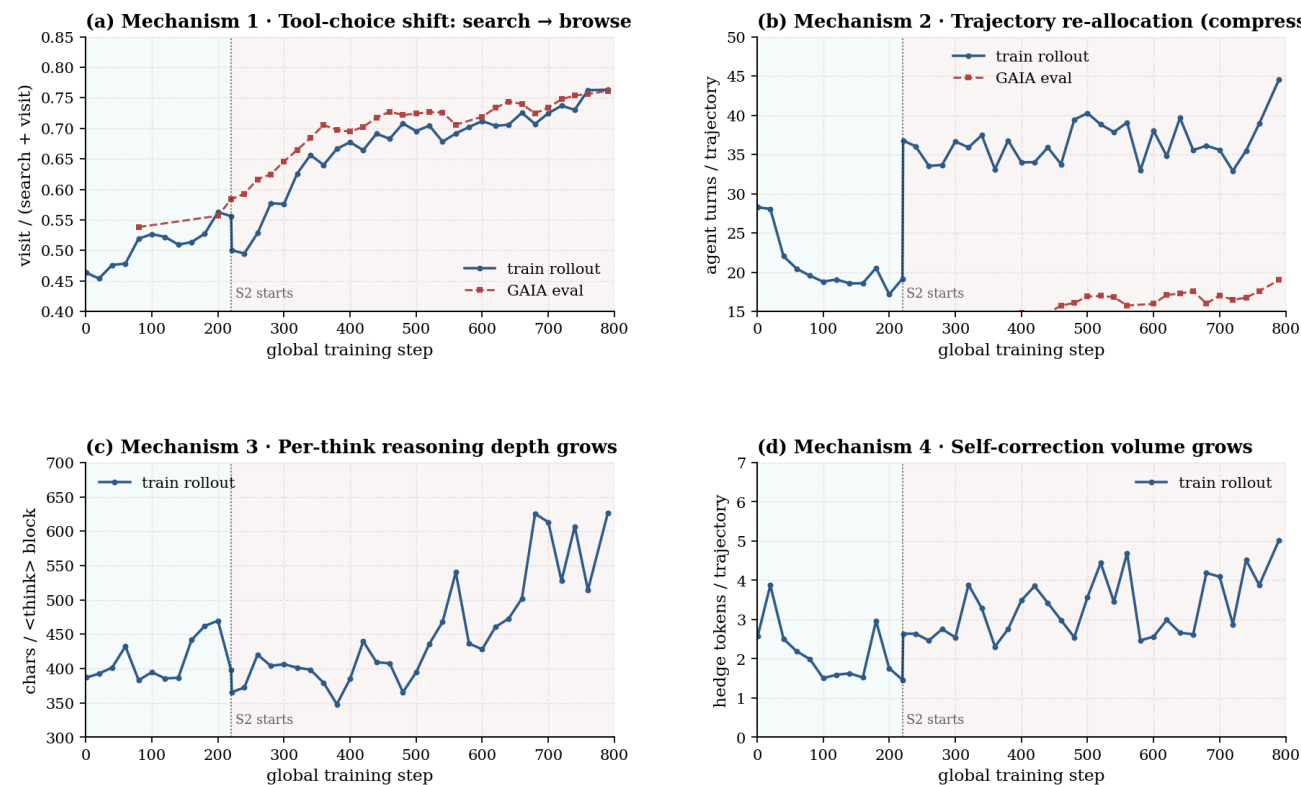
## OBSERVATION 3

pass@4 saturates earlier than pass@1 — capability is there from day one; what improves is *consistency*.

# Four mechanisms — together, they explain the gain.

Two are structural ( $M1 \cdot M2$ ); two emerge as their downstream effect ( $M3 \cdot M4$ ).

Four behavioral mechanisms underlying the Sim2Real gain (S1 0-220, S2 0-570 spliced at step 220)



Blue = train rollout (300 traj/step). Red = GAIA eval (412 tasks). Panel (d) is train-only because GAIA logs strip <think> blocks.

Per-mechanism shifts across checkpoints; vertical dashed line = S1 → S2 splice.

## M1 Tool-choice shift

search → browse · 0.46 → 0.76

## M2 Trajectory re-allocation

turns · 28 → 19 → 45 (U-shape)

## M3 Per-think depth

chars / think · 387 → 626

## M4 Self-correction volume

hedges / traj · 1.5 → 5.0

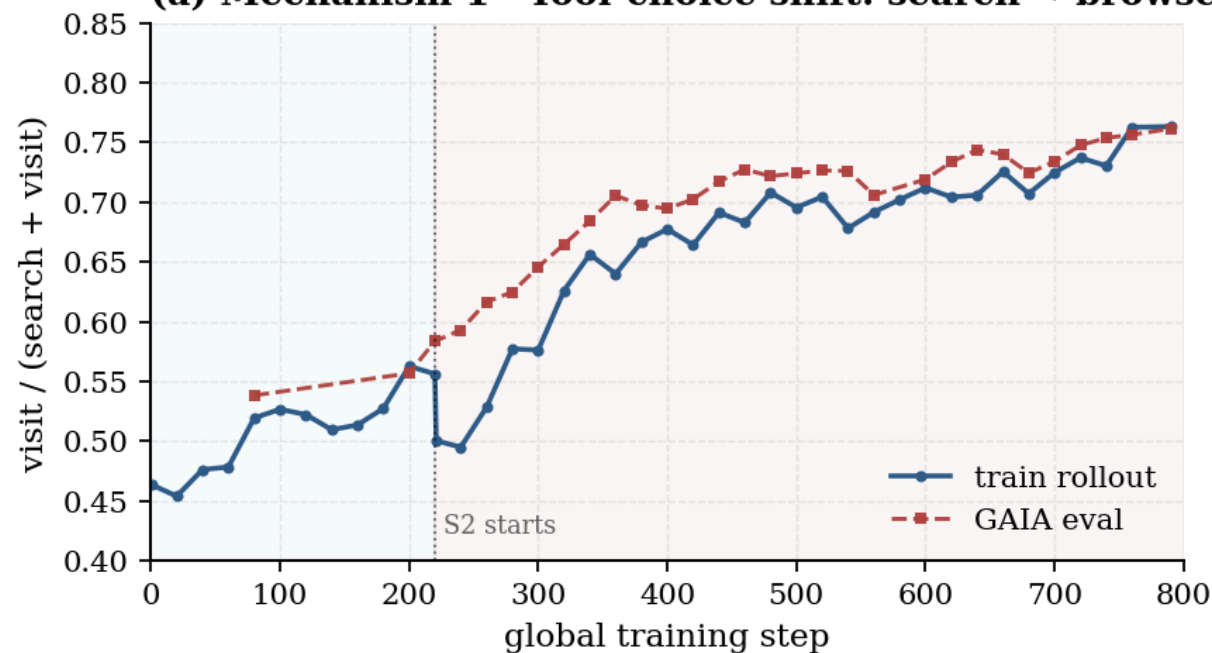
## READING ORDER

We unfold M1 → M4 on the next four slides.

# From keyword search to deep browsing.

*The policy stops treating search as the answer — it uses search as a pointer toward evidence.*

(a) Mechanism 1 · Tool-choice shift: search → browse



*browse / (search + browse) per trajectory across checkpoints. Shaded region = S1; vertical line = splice.*

browse / (search + browse)

|      |     |      |
|------|-----|------|
| step | 0   | 0.46 |
| step | 220 | 0.56 |
| step | 460 | 0.72 |
| step | 790 | 0.76 |

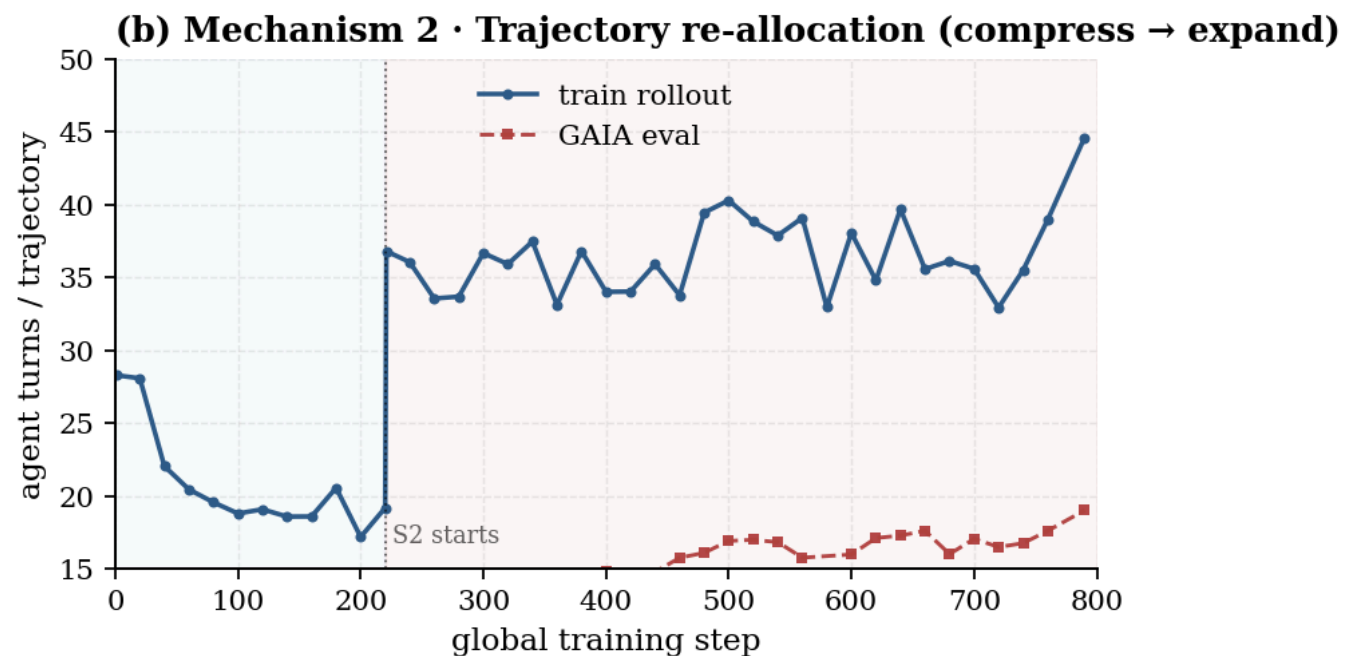
## INTERPRETATION

Search becomes a **pointer**, not an answer. Hits get **opened and read** instead of paraphrased.



# Compress first — then expand into harder problems.

A U-shape: the policy learns efficiency in S1, then re-invests its budget in depth during S2.



agent turns / trajectory — compress during S1 (28 → 19), expand during S2 (19 → 45).

turns / trajectory

28 → 19 → 45

S1 start · S1 end · S2 end

PHASE A · S1 (0 → 220)

Policy learns to **compress** — fewer wasted turns per task.

PHASE B · S2 (220 → 790)

Harder tasks + 48k context — the policy **re-invests** its saved budget into deeper exploration.

# Each reasoning block carries more content.

*The policy reasons more carefully per <think> — not more often per turn.*

chars / <think> block

**387 → 626**

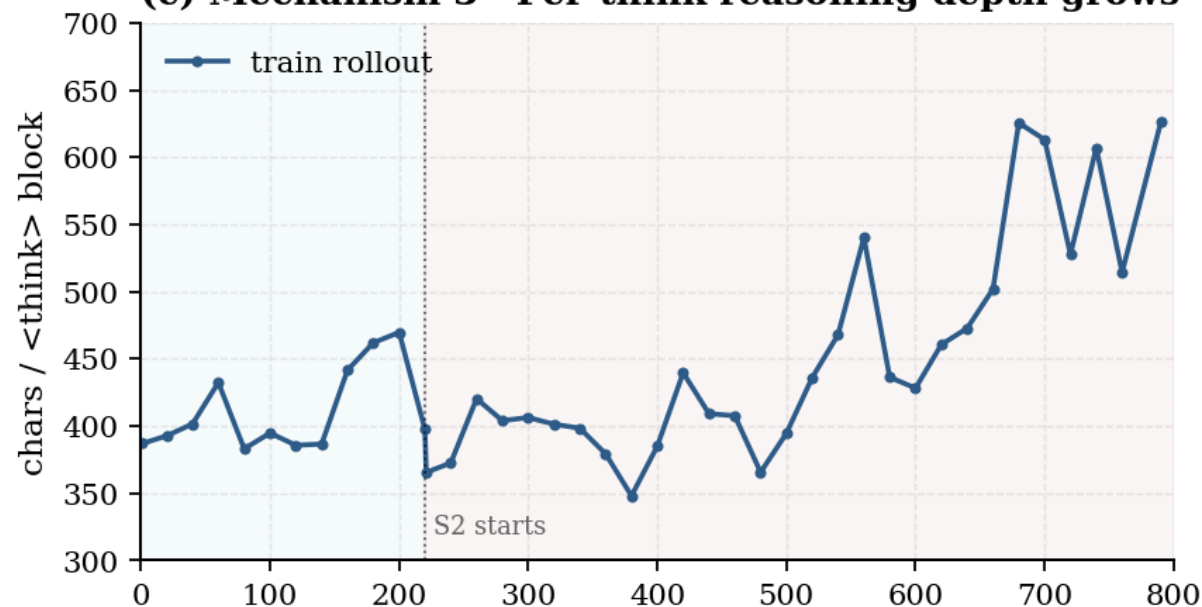
*≈ + 62 % per reasoning block*

WHY DEPTH — NOT FREQUENCY?

Number of <think> blocks per turn stays roughly flat.  
What grows is the **content density** inside each block.

*The policy reasons more carefully — not more often.*

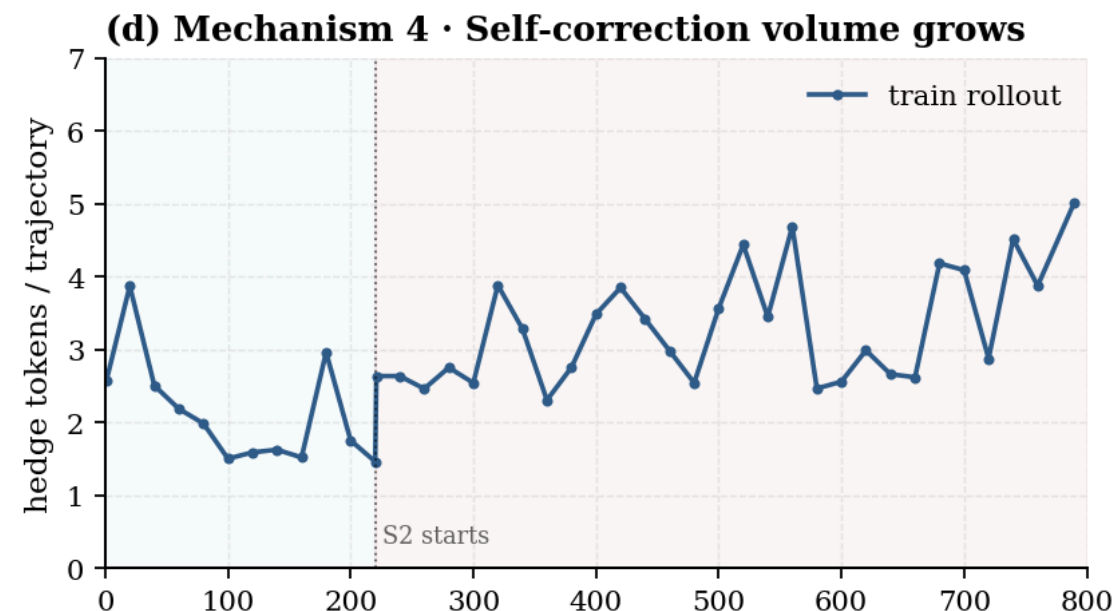
(c) Mechanism 3 · Per-think reasoning depth grows



Average characters per <think> block; 387 at the start of S1, 626 by the end of S2.

# More hedging and verification moves per trajectory.

A volume effect — the trajectory just gets longer and contains more checking.



Hedge tokens per trajectory — "wait" / "verify" / "actually" / "let me check".

hedge tokens / trajectory

**1.5 → 5.0**

$\approx 3.3 \times$  more verification moves

## HONEST FRAMING

Hedges per token barely changes — the trajectory simply gets longer and contains more verification.

*Self-correction is a volume effect, driven by M2's expanded budget.*

# What we ruled out — and the resulting causal chain.

Six plausible signals failed the monotonicity test. Four survived — and form a single chain.

RULED OUT BY THE DATA

| CANDIDATE SIGNAL          | WHY IT FAILED                              |
|---------------------------|--------------------------------------------|
| Answer-length growth      | Confounded with task difficulty in S2      |
| Tool calls per turn       | Stable across both stages — no shift       |
| Think frequency per turn  | Flat — depth, not count, grows             |
| Hedge density (per token) | Almost unchanged — only volume grows       |
| Vocabulary diversity      | No monotone trend across checkpoints       |
| Error / retry rate        | Drops in S1, rebounds in S2 — non-monotone |

SURVIVING CAUSAL CHAIN



# Three things to remember.

*If the deck is forgotten, the following three claims still stand on their own.*

## 01

### **Local environments are enough.**

A 32 M-page mirror + a lightweight browse tool replaces \$59 K–\$243 K of commercial APIs, delivers 10×–46× speed-ups, and unlocks the trajectory volume needed for on-policy RL at scale.

## 02

### **Simple-but-scalable beats clever-but-fragile.**

Atomic-capability synthesis, on-policy GRPO, and a 2-stage curriculum — each individually familiar — combine to lift a 4 B model past 30 B teachers and most commercial systems.

## 03

### **The gain has a behavioral signature.**

Two structural mechanisms (tool-choice shift, trajectory re-allocation) act first; depth and self-correction emerge as their downstream effect. Sim2Real is sustained because the behavior — not just the score — transfers.

# Thank you.

*Questions & discussion welcome.*

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PAPER

[arxiv.org/abs/2604.17931](https://arxiv.org/abs/2604.17931)

CODE

[github.com/Wanli-Lee/Agentic\\_RL](https://github.com/Wanli-Lee/Agentic_RL)

MODEL

[hf.co/litererresearcher-4b](https://hf.co/litererresearcher-4b)

*Wanli Lee · with thanks to all co-authors, reviewers, and the infrastructure team.*